Republic of the Philippines
University of Southeastern Philippines
Iñigo St., Obrero, Davao City 8000
COLLEGE OF ENGINEERING

PROGRAM STRUCTURE

DEGREE PROGRAM: Bachelor of Science in Mechanical Engineering

Current CMO Guiding Program:

The guiding principles followed in the formulation of the policies, standards and guidelines (PSG) of CMO 97, series of 2017, are the pertinent provisions of Republic Act (RA) No. 7722, otherwise known as the "Higher Education Act of 1994," in pursuance of an outcomes-based quality assurance system as advocated under CMO 46 s. 2012 (Policy-Standard to Enhance Quality Assurance (QA) in Philippine Higher Education through an Outcomes-Based and Typology-Based Quality Assurance) and as addendum to CMO 37, s. of 2012 (Establishment of an Outcomes-Based Educational (OBE) System in Higher Education Institutions (HEIs) offering Engineering Programs); and as adopted by the Commission en banc through Resolution No. 788-2017 dated October 24, 2017. This PSG implements shift to outcomes-based education leading to competency based standards. It specifies the 'core competencies' expected of BS Mechanical Engineering graduates "regardless of the type of HEI they graduate from." However, in recognition of OBE and the typology of HEIs, this PSG also provide ample space for HEIs to innovate in the curriculum in line with the assessment of how best to achieve learning outcomes in their particular contexts and their respective missions.

Program Description:

Mechanical Engineering is a profession that concerns itself with mechanical design, energy conversion, fuel and combustion technologies, heat transfer, materials, noise control and acoustics, manufacturing processes, rail transportation, automatic control, product safety and reliability, solar energy, and technological impacts to society. Mechanical engineers study the behavior of materials when forces are applied to them, such as the motion of solids, liquids, gases, and heating and cooling of object and machines. Using these basic building blocks, engineers design space vehicles, computers, power plants, intelligent machines and robots, automobiles, trains, airplanes, furnaces, and air-conditioners. Mechanical engineers work on jet engine design, submarines, hot air balloons, textiles and new materials, medical and hospital equipment, and refrigerators and other home appliances. Anything that is mechanical or must interact with another machine or human being is within the broad scope of mechanical engineering.

VISION

A PREMIER UNIVERSITY IN THE ASEAN REGION

MISSION

USeP shall produce world-class graduates and relevant research and extension through quality education and sustainable resource management.

CORE VALUES
USeP is a community of scholars that values:
U – Unity
S – Stewardship
E – Excellence
P – Professionalism

Program Educational Objectives		Mission-Vision and Core Value Alignment						Reference Source or CMO Provision Compliance
		Vision	Mission	Core Value				
				U	S	E	P	
PEO1	Demonstrate expertise in the field of Mechanical Engineering	√	√		√	√	√	USEP Brochure, CMO 97, s. 2017
PEO2	Actively participate in professional communities to nourish ever-developing careers	√	√	√	√		√	USEP Brochure, CMO 97, s. 2017
PEO3	Demonstrate gender sensitive leadership and commitment to work in diverse cultural and interdisciplinary backgrounds	√	√	√		√	√	USEP Brochure, CMO 97, s. 2017
PEO4	Exhibit globally competitive skills	√	√		√	√	√	USEP Brochure, CMO 97, s. 2017

PERFORMANCE INDICATORS

Complete the performance indicators for each of the outcomes that we are able to generate during the workshop. The starting s tatements are not yet complete; they are just starting phrases to guide you in your performance indicators formulation

Institutional Graduate Attributes:

IGA1 – Critical Analysis
IGA2 – Leadership and Teamwork
IGA3 – Service-Oriented
IGA4 – Professional Competence
IGA5 – Communication
IGA6 – Life-Long Learning

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Institutional Graduate Attributes:

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Program Attribute	Program Outcomes	Performance Indicators
IGA1	**a** Apply knowledge of mathematics and sciences to solve engineering problems;	1 Apply concepts in mathematics to solve engineering problems
IGA4		2 Apply principles in chemical sciences to solve engineering problems.
IGA6		3 Apply principles in physical sciences to solve engineering problems
IGA1, IGA4	**b** Design and conduct experiments, as well as to analyze and interpret data	1 Perform laboratory experiments as prescribed by standard procedures.
IGA5, IGA6		2 Design an experiment to achieve a specific objective
IGA1, IGA2	**c** Design a system, component, or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability, in accordance with standards.	1 Develop an engineering solution that have various constraints as factors
IGA3, IGA4		2 Design a system or process based on various constraints
IGA5, IGA6		
IGA2	**d** Function on multidisciplinary teams	1 Shows tact in dealing with people he/she comes in contact with.
IGA3		2 Capable of learning from and listening to co-members
IGA6		3 Shows respect and courtesy in dealing with peers and superiors.
IGA1	**e** Identify, formulate, and solve engineering problems	1 Interpret statement problems into mathematical expressions.
IGA4		2 Apply concepts, theories, and principles in solving engineering problem(s)
IGA6		3 Solve specific engineering problem related to discipline.
IGA4	**f** Understand professional, social and ethical responsibility	1 Internalizes the professional code of ethics for specific discipline.
IGA5		2 Applies the principles of professional ethical standard and social responsibility to actual practice
IGA6		
IGA4	**g** Communicate effectively	1 Articulate the convention and style in oral communication that conveys complete thought and information.
IGA5		2 Articulate the convention and style in written communication that conveys complete thought and information.
IGA6		3 Demonstrate proficiency in using graphs, tables, drawings, symbols, etc. applicable in the discipline.
IGA1, IGA3	**h** Understand the impact of engineering solutions in a global, economic, environmental, & societal context through broad education	1 Relate engineering solutions to global, economic, environmental, and societal context.
IGA4, IGA6		2 Describe engineering solutions in application to global, economic, environmental, and societal context.
IGA1, IGA2	**i** Recognize the need for, and an ability to engage in life-long learning	1 Demonstrate ability to learn independently.
IGA3, IGA4		2 Shows appreciation and understanding of the engineering profession.
IGA5, IGA6		3 Understand the linkage or connection between his/her task to previous, intervening and subsequent tasks.
IGA2, IGA3	**j** Apply knowledge of contemporary issues	1 Identify the socio-economic, political, and environmental issues related to the practice of engineering
IGA4, IGA5		2 Explain the possible impact of technology to local, national, and international issues
IGA6		3 Shows leadership skills in managing a multidisciplinary group.

IGA1	**k**	Use techniques, skills, and modern engineering tools necessary for engineering practice	1	Handles the details of the work assigned to him/her.
IGA4			2	Demonstrate the ability to operate the equipment needed in the engineering practice.
IGA6			3	Use modern technology in performing engineering practice.
IGA1, IGA2	**l**	Apply knowledge of engineering and management principles as a member and leader in a team, and to manage projects and in multidisciplinary environments	1	Capable of learning from and listening to co-members.
IGA3, IGA4			2	Shows leadership skills in managing a multidisciplinary group.
IGA5, IGA6			3	Participates in discussions and comes up with sound suggestions to problems.

There are about six (6) fundamental attributes for any professional that belongs to high-capabilities being developed in Higher Education. To check if your program outcomes are complete, you must have a representative outcome to each of the attributes indicated above. The attributes above are minimum; you can add additional attributes and outcomes. It is possible that in one attribute, there are more than one outcome

CURRICULUM MAP

List courses such that first year first semester courses appear first, until the last semester of the last year of the degree program. Exclude in the list the following: (a) General Education courses (b) Elective courses. As much as possible, observe the following:

LEGEND

Code	Descriptor	Definition
I	Introductory Course to an Outcome	A formative course to an outcome
E	Enabling Course to an Outcome	A course strengthening further an outcome
D	Demonstrative Course to an Outcome	A course demonstrating an outcome

Year Level	Semester	Units	Courses in the Curriculum	Program Outcomes											
				a	b	c	d	e	f	g	H	I	J	K	l
	First	3	Purposive Communication												
		3	Mathematics in the Modern World												
		3	Understanding the Self												
		3	Engineering Calculus 1	I				I							
		4	Chemistry for Engineers	I	I			I							
		2	Orientation to Mechanical Engineering				I		I	I	I				
		1	Engineering Drawing and Plans				I			I					
		2	Movement Enhancement												
		3	National Service Training Program 1												
	Second	3	Ethics												
		3	Science, Technology, and Society												
		3	Arts Appreciation												
		3	Engineering Calculus 2	E				E							
		4	Physics for Engineers	E	I		I	E							
		3	Computer Fundamentals & Programming		I							I		I	
		2	Fitness Exercises												
		3	National Service Training Program 2												
Year II	First	3	The Life and Works of Jose Rizal												
		3	Readings in Philippine History												
		3	Differential Equations	E				E							
		3	Engineering Data Analysis	I				I			I				

		3	Thermodynamics 1	I				I							
		1	Workshop Theory and Practice			I								I	
		2	Basic Electrical Engineering	E				E							
		1	Basic Electrical Engineering Laboratory		E									E	
		5	Mechanics of Rigid Bodies	E				E							
		2	Physical Activities Towards Health & Fitness I												
	Second	3	Advanced Mathematics for Mechanical Engineering	E				E							
		3	Thermodynamics 2	E				E							
		2	Machine Elements	E				E							
		1	Machine Elements Laboratory				I								
		2	Machine Shop Theory			E								E	
		2	Heat Transfer	E				E							
		2	Basic Electronics	E				E							
		1	Basic Electronics Lab		E									E	
		1	Computer-aided Drafting						I	I		I		I	
		4	Mechanics of Deformable Bodies	E				E							
		3	Fluid Mechanics	E				E							
		2	Physical Activities Towards Health & Fitness II												
Year III	First	3	The Contemporary World												
		3	General Education Elective 1												
		3	Fluid Machineries	E				E							
		1	Mechanical Engineering Laboratory 1		I		I			I					
		3	Refrigeration Systems	E		E		E							
		2	Combustion Engineering	E				E							
		1	Computer Applications for ME	E										E	
		2	DC & AC Machinery	E				E						E	
		1	DC & AC Machinery Laboratory		E		E								
		3	Engineering Economics	E					I		I		I		I
		2	Material Science and Engineering for ME	E	E			E							
		1	Material Science and Engineering for ME Lab		E		E								
	Second	3	General Education Elective 2												
		3	General Education Elective 3												
		3	Power Plant Design with Renewable Energy	D		D		D			D				
		1	Power Plant Design with Renewable Energy Laboratory				E								
		3	Machine Design 1	D		D		D							
		2	Control Engineering	E				E			E				
		1	Control Engineering Laboratory		E			E							E
		2	Mechanical Engineering Laboratory 2		E		E			E					
		3	Air-conditioning and Ventilation Systems	E		E		E							
		3	Methods of Research						E	E	E		E		E
		3	Basic Occupational Safety and Health					E	E	E			E		
Year IV	First	3	ME Project Study (Thesis)*	E	E	E	E		E	E				E	
		2	Machine Design 2	D		D		D							
		1	Machine Design 2 Laboratory	D		D		D							
		3	Industrial Plant Engineering	D		D		D			D				

		1	Industrial Plant Engineering Laboratory				E								
		2	Mechanical Engineering Laboratory 3			D		D		D					
		2	Industrial Processes w/ Plant Visits & Seminars	E				E			E				
		2	ME Laws, Ethics, Contracts, Codes and Standards						E	E			E		
		2	Vibration Engineering	E				E							
		2	Mechanical Engineering Elective 1 (Correlation Course 1)	E		E		E						E	
		2	Engineering Management				E		E	E	E		E		E
		3	Technopreneurship				E		E	E					E
	Second	3	ME Project Study (Thesis)	D	D	D	D		D	D					
		2	OJT (320 hours)	E				E			E				
		2	Mechanical Engineering Elective 2 (Correlation Course 2)	E		E		E						E	
		1	Mechanical Engineering Elective 3	E				E			E			E	

REFLECTION FORM

After going through this exercise of planning for your course, give yourself a 15-minute break and reflect on the following questions below. For each of the question, indicate the rating

	5	4	3	2	1
Do you think that your course now has a better purpose to the professional or personal life of the student?	√				
Do you think that your course now has a better alignment to the intentions of the degree program or the intentions of the institution?	√				
Do you think that your course planning technique has gone better this time than before?	√				
Do you think you have designed a course now with a better plan as to how students will progress inside the course?	√				
Do you think your topics will be better anchored to a specific purpose/s than before?	√				
Do you think you are planning your courses now for students to succeed and achieve capabilities?	√				
Do you think that if this course will be properly implemented, your students will be developed better?	√				
Do you think that if you are given training on better instructional delivery, this course is exciting to teach?	√				

Legend

5 Very Likely
4 Less Likely
3 Cannot tell
2 Not Likely
1 Definitely Not

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